

Self-Supervised 3D Reconstruction and Ego-Motion Estimation via On-Board Monocular Video

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Abstract—Recovering the three-dimensional structure information from a monocular camera is significant for automated driving, robot navigation, and traffic safety assessment. Recent work has solved various tight issues on self-supervised monocular depth estimation of leveraging on-board videos, such as occlusion/disocclusion, dynamic objects, and scale inconsistent. Nevertheless, rare work focuses on the model's prediction confidence and underlying relations between depths, while they are essential for a decision-making system and performance improvement, respectively. This paper proposes a novel scheme, that of correlation-aware structure, to dig into the relations between depths, converting the independent depths into a graph-like connected depth map. Subsequently, a Gaussian estimator is devised to predict the depth map and uncertainty map concurrently. The uncertainty map can show us problematic regions where it is difficult to predict, from which we further develop uncertainty-based strategies to improve the performance. Specifically, we propose a simple image preprocessing method to overcome the gradient locality issue caused by low-texture, especially in smooth roads and shadows. Also, to avoid the influence of high-uncertainty regions, we propose a solidity-aware mask to recognize the reliable pixels for training in the image. The experiments on the KITTI dataset show that our method results in a competitive performance in both depth and ego-motion estimation tasks compared with the state-of-the-art methods. Besides, additional experiments on the Make3D and Cityscapes datasets demonstrate our method's strong generalization capability and practicality.

Index Terms—3D reconstruction, monocular depth estimation, perception, uncertainty estimation, visual odometry.

I. INTRODUCTION

AUTONOMOUS driving is currently a heated research topic, which can significantly improve traffic efficiency and safety, protecting people's property and life. Driving environment perception is crucial for a self-driving vehicle, especially in three-dimensional (3D) structure perception, which is heavily correlated to final decision-making. A wrong or insufficient decision might result in unexpected consequences,

even life and property loss. Accordingly, accurately perceiving the 3D driving environment of an automated vehicle comes to be necessary. Besides, traffic safety assessment also requires good knowledge of the scene's 3D structure to locate the objects' exact 3D positions.

At present, autonomous vehicles generally adopt Light Detection and Ranging (LiDAR) to obtain 3D information. However, LiDAR is too expensive to be widely used. And, LiDAR cannot acquire a complete 3D structure of the scene, as the scanning range is limited. The LiDAR sensor is also heavily affected by bad weather, such as snow, rain, sand, and dust. Compared with LiDAR, the camera is much cheaper and commonly used. Simultaneously, we expect that our 3D reconstruction method can be used in autonomous driving and other transportation situations, such as traffic safety assessment and traffic surveillance video processing. Therefore, we focus on self-supervised 3D reconstruction based on monocular camera data, such as the KITTI, Make3D and Cityscapes datasets [1]–[3], considering efficiency, economy, and universality. Even though a self-supervised method solely can recover a relative depth map temporally, the reconstructed object shape information can effectively benefit other computer vision tasks, such as object detection and semantic segmentation. Additionally, we can easily obtain the actual scale of depth with the aid of the vehicle's velocity.

To the best of our knowledge, Zhou *et al.* [4] are the first to utilize monocular video to reconstruct the scene's 3D structure. Following that, there is a large amount of research involved in self-supervised 3D reconstruction via monocular video, focusing on solving a set of inherent issues of this pipeline, including modeling motion objects, occlusion/disocclusion, scale inconsistency. The 3D reconstruction performance gap between supervised and self-supervised methods comes to be smaller and smaller through these excellent works. Notably, Bian *et al.* [5] proposed a simple but effective scale consistency network, which means that we can utilize a single scale factor to recover the actual 3D structure of all frames in a video sequence, significantly raising the probability of being a practical application.

Nevertheless, some inherent issues, such as reflective, far, and low-texture regions, remain open. Moreover, these difficult regions regularly appear in the driving environment, such as the car windows (reflective areas), shadows, and roads (low-texture regions). Additionally, uncertainty estimation is significant for a practical application system, especially for a self-driving system that requires absolute security. Intuitively, high uncertainty indicates that some implicit or explicit issues

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exist, inhibiting the network training. Accordingly, devising some special modules in solving the problems mentioned earlier will significantly benefit performance improvement of self-supervised 3D reconstruction, promoting its application in a real-world system.

To the best of our knowledge, none of the previous work takes the aforementioned problems into account. To bridge these gaps, we propose:

- A correlation-aware structure to model the relations between depths. By doing that, the depth map with independent depths originally is converted into a graph-like depth map, contributing to solving reflective and far regions.
- A Gaussian estimator to predict the depth map and ego-motion with their uncertainties, demonstrating the problematic regions and promoting performance gain as a byproduct.
- A simple preprocessing method to handle the gradient locality issue from low-texture, particularly for smooth roads and shadows.
- A solidity-aware mask to discover the reliable pixels in projection, thereby mitigating high-uncertainty regions' influence.
- A fusion scaling method to tradeoff different preferences, further obtaining performance improvement.

To evaluate the effectiveness of our method, we organize the paper as follows. Foremost, we have a short literature study in the next section. Then, a mathematical description comes to define the proposed problem. Afterward, the method that we intend to adopt is introduced. Following that, exhaustive experiments demonstrate our model's superiorities. Ultimately, a conclusion section comes to summarize the paper. Also, we illustrate some failure cases, further discussing the model's limitations.

II. RELATED WORK

Recovering the scene's 3D structure and camera ego-motion from the video is crucial for autonomous driving, robot navigation, and traffic safety, drawing many researchers' attention.

A. Supervised Depth Estimation

In a supervised manner, the hand-crafted features are first utilized for estimation [6], [7]. Specifically, Saxena *et al.* predicted the depth for local regions instead of estimating depth for each pixel by combining absolute depth features extracting from textures and gradients of image and relative depth features from the image's histograms. Eventually, Markov random field (MRF) has been used to obtain a comprehensive estimation. Following that, many works have been done to explore the image's monocular cues, then predicted the depth map [8]–[11].

Traditional approaches are limited to extract some advanced and abstract features from imagery compared with convolutional neural network (CNN). To that end, Eigen *et al.* have done some pioneer-works in estimating depth from a single image using CNN [12]. Since then, quite a few works have been done to improve the performance [13]–[16], mainly by

devising novel CNN structures. Of note is that the CNNs merely can extract local features of the imagery. However, global information also contributes to depth estimation. Consequently, the researchers adopted two ways to solve this problem: one of which depends on the conditional random fields (CRFs) to obtain the global information [12], [17]–[22], while another solves that with the aid of the recurrent neural network (RNN) [23]–[28]. Simultaneously, multi-task learning has been used to improve the depth estimation performance as well [12], [20].

Additionally, it is worth mentioning that depth estimation not only can be cast into a regression problem but also a classification problem [17], [29]. More specifically, the researchers first discretized the continuous depth as many bins, subsequently allocating a specific label for each bin. In that case, the regression problem is converted into a classification problem.

B. Self/Un-Supervised Depth Estimation

A well-known issue of supervised learning is that tremendous labeled training data is required, while it is way too vulnerable due to the noises from the environment and sensor itself. In contrast, a self-supervised way can capture ego-motion and depth from monocular video without any other sensors, by supervision of warping-based view synthesis [4]. Since then, many works have been done in this field [4], [30]–[35].

Representatively, Godard *et al.* focused on handling the occlusion/disocclusion issue [36]. More specifically, the minimal one among the reprojection losses from different reference images will be regarded as the final loss. Additionally, the moving objects in the scene are out of the reprojection as we assume that no motion objects exist. Accordingly, Casser *et al.* proposed that potential motion objects can be masked out using the advanced instance segmentation technology, then predicting the ego-motion and object motion separately [37].

As for gradient locality caused by low-texture, quite a few works [31], [38] have followed Zhou's work [4] using multi-scale training to solve it. However, this sort of pipeline requires extra-computation during training, and it cannot help as expected in terms of Bian's work [5] either. Another issue is that the depth and ego-motion from self-supervised methods are ambiguous in scale. To that end, Bian *et al.* proposed geometry consistency loss as supervision to encourage the model to obtain the scale-consistent depth and ego-motion within a continuous sequence [5].

The model's efficiency is vital for a real-world application. Thus, Park and Yang *et al.* deployed sensor fusion technologies, with the assistance of lidar, stereo fusion, and visual odometry, to obtain more accurate and faster depth estimation results [39], [40]. Not limited to monocular, stereo methods have also been developed [38], [41]. The generative adversarial network (GAN) performs pretty well on depth estimation as well [23], [42].

Existing work did not model the relations between depth and prediction uncertainties. They did not consider how to mitigate the influence of high-uncertainty regions, either. In this paper,

we will bridge these gaps by proposing a set of specialized strategies.

III. PROBLEM SETUP

The model can be divided into two separate parts, in which one is the depth network denoted as Ψ_D while another is the ego-motion network marked as Ψ_E . The depth network's input is target imagery $I \in \mathbb{R}^{h \times w \times c}$, where h , w , and c represent the height, width, and the number of the channel of the image, respectively. Furthermore, we can describe the depth network as $\Psi_D : I \in \mathbb{R}^{h \times w \times c} \rightarrow D \in \mathbb{R}^{h \times w}$. The pose network's input is the image sequence, writing as $I \in \mathbb{R}^{l \times h \times w \times c}$ that satisfies the target image located in the middle position of the sequence, and l is the length of the sequence. In this work, we follow Zhou's work [4], setting the sequence's length as 3. Specifically, $\Psi_E : I \in \mathbb{R}^{3 \times h \times w \times c} \rightarrow T \in \mathbb{R}^6$.

The image sequence can be denoted as $I = (I_r^1, I_t, I_r^2)$, and the 6-degree pose vector T can be further converted into transform matrix $E_{I_t \rightarrow I_r^j}^{3 \times 3}$. In the case of knowing the depth map D_t corresponding target image I_t from Ψ_D and the transform matrices $E_{I_t \rightarrow I_r^1}^{3 \times 3}$, $E_{I_t \rightarrow I_r^2}^{3 \times 3}$ from Ψ_E , we can reconstruct the target image from the reference image according to Eq. 1, in which p_r , K , $E_{t \rightarrow r}$, $D_t(p_t)$, K^{-1} , and p_t stand for the coordinate in the reference image, camera intrinsic matrix ($\mathbb{R}^{3 \times 3}$), transform matrix between the target image and reference image, depth corresponding p_t , inverse matrix of K , and coordinate in the target image, respectively. By doing that, the reference image I_r^i can be projected to the plane of the target image, and denoted as $I_r^{i'}$. The model is trained by leveraging the difference between the target image and the warped image. By doing that, the depth map and ego-motion can be obtained.

$$p_r \sim K E_{t \rightarrow r} D_t(p_t) K^{-1} p_t \quad (1)$$

IV. METHOD

In this section, we first introduce the overall architecture and loss of the proposed method. Subsequently, the specific losses, correlation-aware structure, Gaussian estimator, overcoming gradient locality, solidity-aware mask, and fusion scaling method are introduced.

A. Model Overview

Towards humanity-like inference, we consider the underlying relations between depths since pure CNNs only extract the image's local information, not considering the information correlation. To the best of our knowledge, there are very few researches considering the relation between depths before. Therefore, we propose a correlation-aware structure to bridge this gap, automatically mitigating some well-known issues at the same time, such as reflection, transparency, and occlusion/disocclusion.

Additionally, the uncertainty estimation is essential and meaningful in a practical system, especially in a self-driving system. However, only rare research has been done considering that. To that end, we propose a Gaussian estimator to predict

the depth map and its uncertainty in pixel-wise. Observations coming out of the uncertainty map can further guide the improvement of performance. Furthermore, the uncertainty map can provide more detailed information regarding the depth and ego-motion estimation models, which can help the decision-making system make safer and more reasonable decisions. Both correlation-aware structure and Gaussian estimator are applied into depth network, while pose network merely adopts Gaussian estimator. Fig. 1 illustrates the framework of our model.

Depth estimation network collaborates with pose estimation network during training, yet they can separately work when inference. During training, we adopt structure similarity (SSIM) loss L_{ssim} to correct the structure of warped image coming out of reference image, which concurrently can handle the illumination changes of real-world scenarios better [5], [43]. The photometric loss L_p between the target and warped image as supervision of details can correct the pixels' positions to obtain more accurate depth and ego-motion. Smoothness loss L_{smooth} can help to predict a smooth depth map.

For assessing the prediction's uncertainty, Gaussian depth and ego-motion have been implemented. Thus, L_{ssim} , L_p , and L_{smooth} are rewritten as L_{ssim}^G , L_p^G , L_{smooth}^G , respectively. Also, the implementation of our solidity-aware mask converts L_{ssim}^G , L_p^G into $L_{ssim}^{G,M}$, $L_p^{G,M}$. According to Bian's work [5], the model trained on a single scale performs better than the model trained on multi scales. Following that, the loss function can be formulated as Eq. 2 ultimately:

$$\mathcal{L} = \alpha \mathcal{L}_{ssim}^{G,M} + \beta \mathcal{L}_p^{G,M} + \gamma \mathcal{L}_{smooth}^G \quad (2)$$

B. Losses

1) *Photometric Loss*: A strong assumption is that all surfaces satisfy the Lambertian reflection [44], which supports the photometric constancy between the warped image $I_r^{i'}$ and the target image I_t . Many previous works successfully trained self-supervised depth network with photometric loss as supervision [4], [36], [37], which can be formulated as follows (Eq. 3):

$$\mathcal{L}_p = \sum_{i=1}^n \frac{1}{N_i} \|I_r^{i'}(p) - I_t(p)\|, p \in N_i \quad (3)$$

where N_i stands for the number of valid points when projection from the i^{th} reference image to the target image, and n is the number of reference images. We take the robustness into account so that the L_1 norm is chosen for that.

2) *SSIM Loss*: SSIM loss emphasizes the structure similarity and mainly captures the image's edges [43]. Besides, the photometric loss is sensitive to illumination changes in the real-world. Thus, we utilize the SSIM loss stated in Eq. 4 to improve that.

$$\mathcal{L}_{ssim} = \sum_{i=1}^n \frac{1}{N_i} \frac{\|1 - SSIM(I_t(p), I_r^{i'}(p))\|_1}{2}, p \in N_i \quad (4)$$

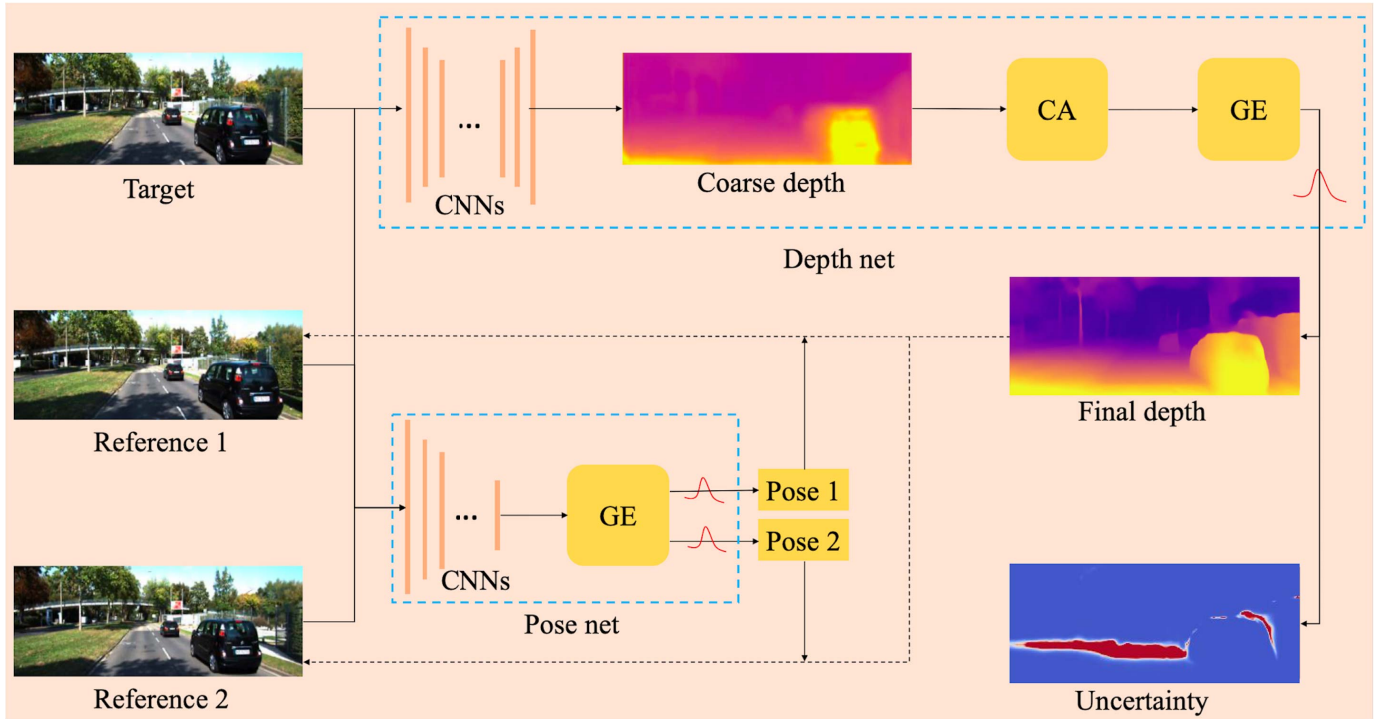


Fig. 1. The architecture of our model. CA stands for the correlation-aware structure, and GE denotes Gaussian estimator. In the depth map, the brighter the color is, the closer the depth is. In the uncertainty map, the redder the color is, the higher the uncertainty is.

3) *Smoothness Loss*: As the difficulty of predicting the depth in low-texture or homogeneous regions of the image, a smoothness prior has been utilized in previous works. Likewise, we adopt edge-aware smoothness loss, following [5], which formulates as Eq. 5 as follows:

$$\mathcal{L}_{smooth} = \sum_p (e^{-\nabla I_t(p)} \cdot \nabla D_t(p))^2 \quad (5)$$

where p belongs to the whole image space, ∇ is the first derivative operator, working along with horizontal and vertical directions, respectively.

C. Correlation-Aware Structure

A critical capability of humanity is that humankind can obtain an accurate depth prediction result by force of the nearby information, even though the situation is quite challenging to estimate. Namely, potential relations between depths can be dug, further improving depth estimation's performance. Specifically, a specified depth should exhibit specific correlations with neighboring depths. Roughly, an image's horizontal depths should have similar values within the same semantic region, and the vertical depths trend to change gradually. This kind of relationship can provide constraints over depth estimation, mitigating some tight problems in self-supervised depth estimation, such as reflection and occlusion.

Prior to obtaining the ultimate depth map, the network first gets a coarse depth map, and it is the input of a correlation-aware structure. In contrast, the correlation-aware structure's output is the depth map of considering the depths'

underlying relations. We can formulate the underlying correlation as Eq. 6.

$$D(i, j) = \sum_{m=0,1,-1} \sum_{n=0,1,-1} w_{i+m,j+n} \kappa(F(i, j), F(i+m, j+n)) \quad \text{s.t. } m \cdot n = 0 \quad (6)$$

$D(i, j)$ stands for the depth on i^{th} row and j^{th} column of the depth map, while $F(i, j)$ is the coarse depth corresponding to $D(i, j)$. $w_{i+m,j+n}$ represents the learnable weight, and $\kappa(x, y)$ is the correlation unit that considers the correlations between x and y . Note that $i+m$ and $j+n$ should be less than the image's height and width concurrently. In this case, $D(i, j)$ is not independent but correlates with nearby depths.

If $D(i, j)$ is correlated with $D(i, j+1)$ and $D(i, j+1)$ is correlated with $D(i, j+2)$, we can infer that $D(i, j)$ is correlated with $D(i, j+2)$. Then, the i^{th} row of the depth map is correlated based on transitivity. Ultimately, each row of the depth map is connected. Similarly, all columns of the depth map will take the correlation between depths into account as well. Therefore, the depth map gets connected globally. Namely, the whole depth map is correlated as a connected graph.

During implementation, we split Eq. 6 into two separate parts based-on the computation directions as Eq. 7, 8, and 9 showed, where $D(i, j)^h$ is the horizontal correlation in opposition to $D(i, j)^v$ that is the vertical correlation. Through that, the problem is converted into digging the inner relations of sequence. To tackle that, we utilize some famous structures as our correlation-aware units to extract the relations between depths, including the basic RNN unit, long-short term memory (LSTM) [45], and gated recurrent unit (GRU) [46]. These units have various capabilities, especially in modeling nonlinear

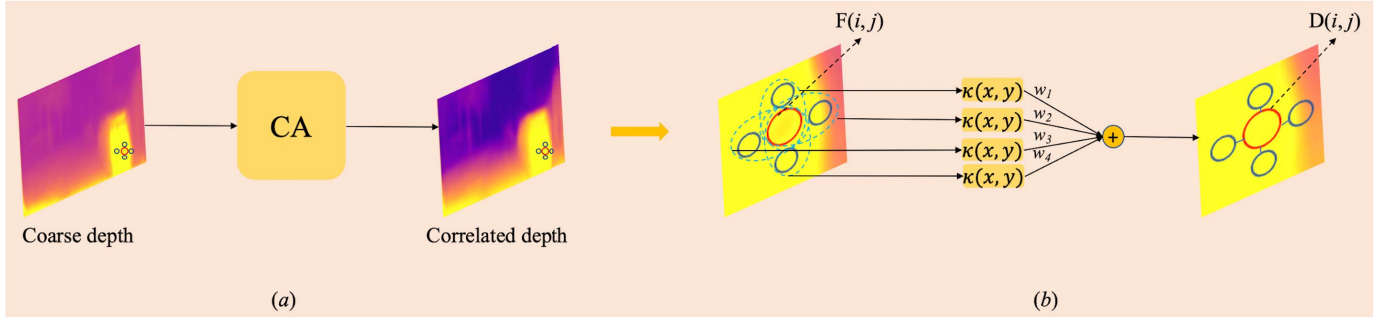


Fig. 2. The proposed correlation-aware structure. CA stands for our correlation-aware structure. (a) demonstrates the visual effect. (b) illustrates the working principle.

and long-term dependency relations. Ultimately, GRU [46] is chosen as our correlation-aware unit. Please kindly find more detailed ablation studies in part C of Section V. Of note is that for simplicity, weight-sharing is adopted for w over different positions of the depth map.

$$D(i, j) = D(i, j)^h + D(i, j)^v \quad (7)$$

$$D(i, j)^h = \sum_{n=1, -1} w_{i, j+n} \kappa(F(i, j), F(i, j+n)) \quad (8)$$

$$D(i, j)^v = \sum_{m=1, -1} w_{i+m, j} \kappa(F(i, j), F(i+m, j)) \quad (9)$$

As Figure 2 (a) depicted, our correlation-aware structure can consider the underlying relations between depths and predict a more reasonable depth for each pixel. Figure 2 (b) illustrates how our model works, corresponding to Eq. 6-9. You also can observe that the originally independent depth map has been converted into a connected depth map through our correlation operation. Consequently, some issues can be worked out automatically, such as the reflective and occlusion/disocclusion regions.

However, the proposed correlation-aware structure differs from the classic CRF technique in the following perspectives:

- CRF aims at minimizing the conditional probability to obtain an overall optimal result, resulting in less attention to local parts. In comparison, our method attempts to find the best solution for each pixel, considering nearby values.
- CRF sets constraints on its output with MRF, but no restrictions for the input. Our method directly models the relations between the input and output instead of putting any constraints on input and output.
- Our correlation-aware structure can choose various correlation functions, such as RNN, LSTM, and GRU, to effectively model nonlinear relations and capture any dependencies between the input and output. Our method can also seamlessly be added to any deep neural networks (DNNs) to form an end-to-end network. CRF cannot achieve that.

D. Gaussian Estimator

Generally, a depth/pose estimation network solely outputs a single depth/pose prediction. Intuitively, it is not consistent with humankind's inference process. Human beings can

predict a depth range using some indefinite words, such as "approximate," instead of a single value, particularly for those depths of challenging to predict. Similarly, our Gaussian estimator tried to learn a dynamic depth map as well.

More specifically, for each position of the Gaussian depth map, a Gaussian distribution formulated in Eq. 10 describes the depth distribution at this point. Similarly, every element of the pose vector is a Gaussian distribution as Eq. 11 formulated. Note that the variances of depths are independent. Still, we cast the pose vector as an entirety, which means that all pose vector elements share a collective variance. $\mathcal{N}(\mu, \delta^2)$ stands for Gaussian distribution with expectation μ and variance δ^2 .

$$D^G(p) \sim \mathcal{N}(\mu(p), \delta(p)^2) \quad (10)$$

$$T^G(i) \sim \mathcal{N}(\mu(i), \delta^2), \quad i = 1, 2, \dots, 6 \quad (11)$$

During training, the Monte Carlo sampling is used to obtain the depth map and pose vector for reprojection, which can be formulated as follows (Eq. 12-13):

$$D'(p) = \frac{1}{s} \sum_{i=1}^s \mathcal{M}(\mathcal{N}(\mu(p), \delta(p)^2)) \quad (12)$$

$$T'(i) = \frac{1}{s} \sum_{j=1}^s \mathcal{M}(\mathcal{N}(\mu(i), \delta^2)) \quad (13)$$

where s is the number of sampling and $\mathcal{M}(\cdot)$ stands for the Monte Carlo sampling. To make it differentiable, we adopt the reparameterization technique formulated as Eq. 14 to obtain the samples. To make δ meaningful, we state that $\delta \leftarrow \delta + \varepsilon$, where ε is an infinitesimal.

$$\mathcal{M}(\mathcal{N}(\mu, \delta^2)) \Leftrightarrow \mu + \delta \cdot \mathcal{M}(\mathcal{N}(0, 1)) \quad (14)$$

In the period of inference, we take s as infinity. In this case, $D'(p)$ should converge to the expectation of $\mathcal{N}(\mu(p), \delta(p)^2)$. Namely, Gaussian distributions' expectations and variances represent the predicted results and uncertainties, respectively. Contradicting with the previous work's point-based estimation, it takes all possible depths into account, and the consequence stands for the average prediction of the model. Therefore, it can not only provide the prediction uncertainty but offering performance improvement as a byproduct. Correspondingly, with Gaussian depth and pose, the losses L_{ssim} , L_p , and L_{smooth} are rewritten as L_{ssim}^G , L_p^G , L_{smooth}^G , respectively.

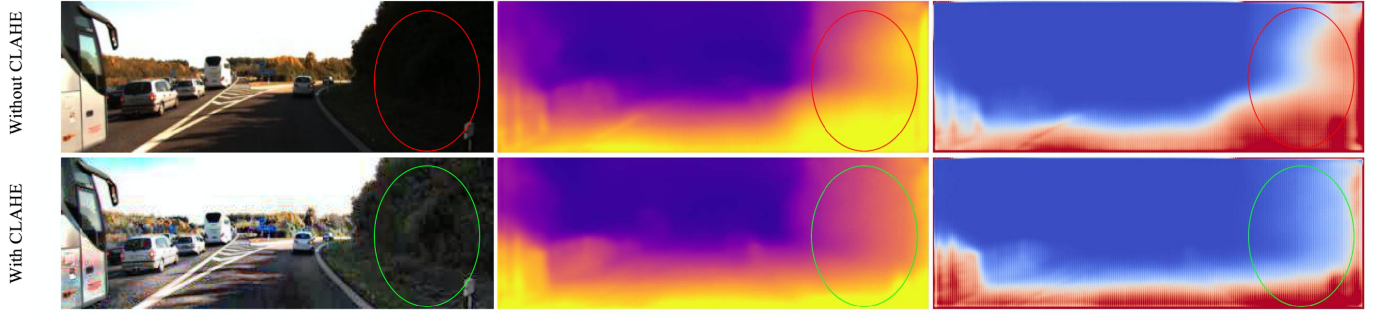


Fig. 3. Visual effects of the CLAHE preprocessing. The first column: RGB images. The second column: depth maps. The third column: uncertainty maps. Issues and improvements are marked with red circles and green circles, respectively.

E. Overcoming Gradient Locality

Some observations from the uncertainty demonstrate a significant issue existing in self-supervised depth and ego-motion estimation, the gradient locality mostly caused by the low textures of the image. For details, many shadows and smooth roads exist in training and testing sets, showing as the high-uncertainty regions according to the uncertainty map. To end that, we propose a simple but efficient data preprocessing method to process the input images for enhancing the pixels' discriminability, especially working for low-texture regions. To achieve that, we propose two ways to do that as follows:

- 1) Converting the RGB color space to HSV color space first, then applying contrast limited adaptive histogram equalization (CLAHE) to V channel merely. Ultimately, transforming processed HSV color space image back to RGB color space image.
- 2) Applying CLAHE to each channel of the RGB image foremost, then overlapping processed channels together.

In our method, we adopt the second strategy to process the images. Fig. 3 illustrates the visual effects of CLAHE preprocessing. It indicates that the CLAHE strategy can significantly improve the gradient locality issue according to depth and uncertainty maps (see more ablation studies in part C of section V).

F. Solidity-Aware Mask

Basically, we assume that the depths with low uncertainties are solid to train the network. In contrast, high-uncertainty predictions are unreliable. In addition, we also assume that problematic regions are difficult to predict, corresponding to high uncertainty. Namely, we can distinguish the challenging areas by uncertainty map. Therefore, paying attention to solid pixels during training can automatically overcome the underlying issues. To mitigate the problematic regions' influence, we propose a self-discovering way to obtain the solidity-aware mask, distinguishing the solid pixels. In Eq. 15, S stands for the solidity-aware mask, while U is the uncertainty map. Particularly $\langle U(i, j) \rangle^K$ represents convolution operation centered with $U(i, j)$, using kernel K , where K is manually devised to compute the mean value within the receptive field. Kernel size is set to 3 in our method (see more ablation studies in

part C of section V).

$$S(i, j) = U(i, j) < \langle U(i, j) \rangle^K \quad (15)$$

By doing that, we can efficiently obtain the solidity-aware mask. Then we incorporate it into valid mask N , which can be formulated as Eq. 16, where $\odot$ stands for element-wise multiplication.

$$N' = N \odot S \quad (16)$$

G. Fusion Scaling

To the best of our knowledge, existing works almost all adopt the scaling method proposed by Zhou's work [4]. However, we find that the proper scaling method also can contribute to performance improvement. And, different scaling ways may have various preferences. We naturally want to propose a kind of scaling method that can take multiple preferences into account. To achieve that, we offer a naive fusion scaling method of merging different scaling approaches. In Eq. 17, D^{gt} denotes the ground truth depths, while D^{pre} is the predicted depths.

$$scale_factor = \frac{1}{2} \left(\frac{median(D^{gt})}{median(D^{pre})} + \frac{mean(D^{gt})}{mean(D^{pre})} \right) \quad (17)$$

V. EXPERIMENTS

In this section, we first introduce the implementation details of our experiments. Then, the comparisons with the state-of-the-art methods are showed. Ultimately, detailed ablation studies demonstrate our model's effectiveness. All experiments are implemented with PyTorch 1.1.0 library and GTX 1080 GPU.

A. Implementation Details

1) *Depth Net*: Our encoder of depth net is ResNet-18 [47] without the last pool layer, flatten operation, and fully connected layer, and we can obtain the encoded features from it. Table I shows the detailed architecture of our decoder. Of note is that we merely use single-scale training in our experiments in terms of Bian's work [5], but the interface of multi-scale training is optional, if necessary.

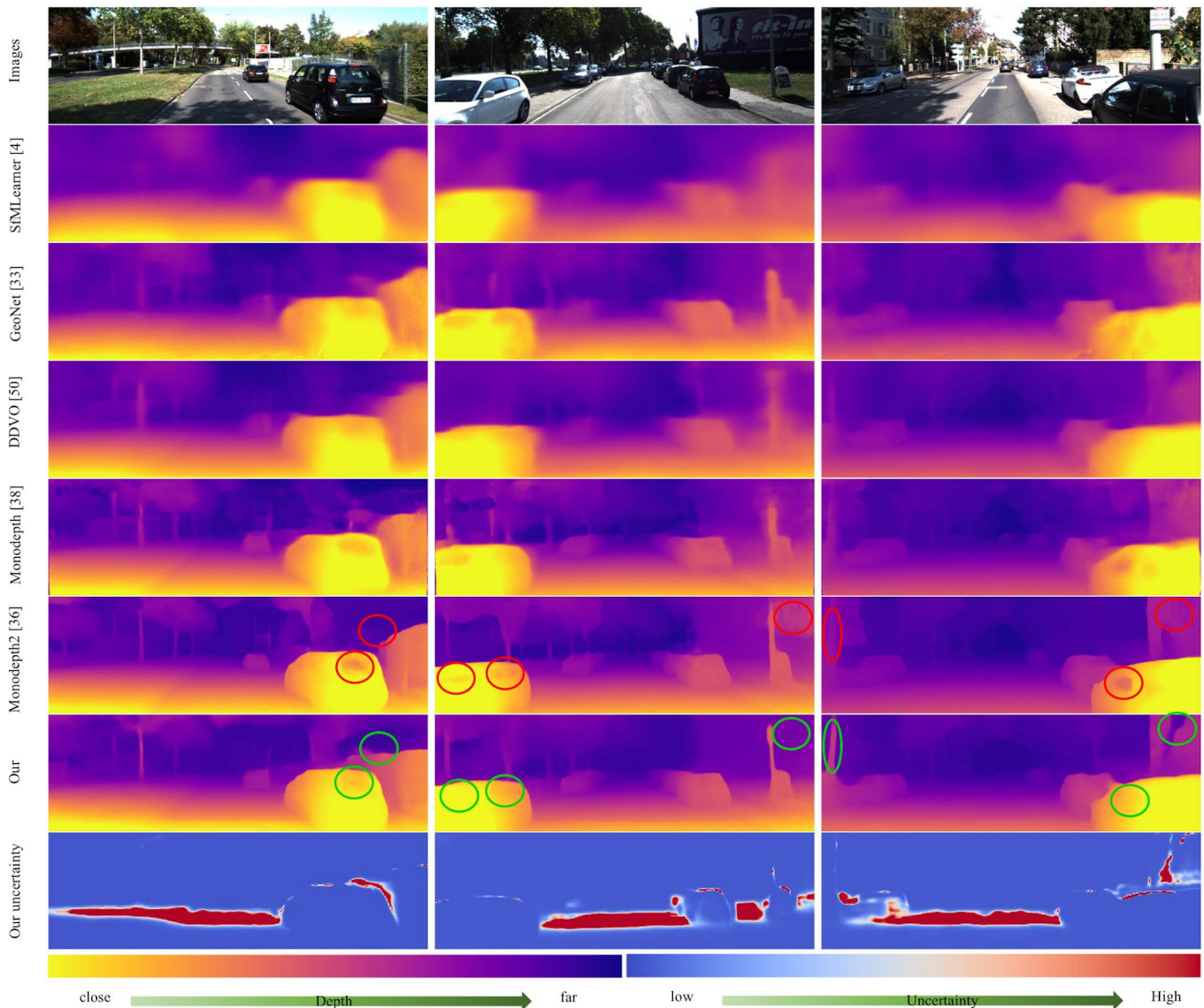


Fig. 4. Qualitative results on the KITTI dataset [1]. Some reflective, far, and smooth regions where it is difficult to predict have been marked with red circles, and our predictions can improve those regions significantly, as shown in green circles. Besides, our uncertainty maps can exhibit high-uncertainty areas, which are problematic regions that are difficult to predict.

2) *Pose Net*: We adopt PoseNet [4] without the mask as our pose net if not specified. The detailed architecture is shown in Table II, and the only difference is that we modify the output channel to 14 instead of 12 in the last pose prediction layer. The extra two channels are uncertainties of poses.

3) *Datasets*: KITTI raw data [1], a large-scale autonomous driving dataset and providing a variety of computer vision task benchmarks, is used for depth estimation. We faithfully follow Zhou's work [4], employing 40,109 and 4,431 sequences for training and validation, respectively. Besides, 697 images from Eigen's split [14] is utilized to test our model.

To demonstrate the generalization capability and practicality, we follow Godard's work [36], directly applying our model trained on KITTI [1] to the Make3D dataset [2] (134 images for testing) unseen during training. Following Godard's work [36], we evaluate on a center crop of 2×1 ratio for Make3D's images. Additionally, we also directly apply

our model trained on KITTI [1] to the Cityscapes dataset [3] without any refinements, exhibiting some qualitative results.

For visual odometry, the KITTI odometry dataset [1] is used. Following Zhan's work [34], we utilize sequence 00-08 and 09-10 for training and testing. Of note is that the depth estimation and visual odometry are trained separately. Our results come from the 3-frame snippets as the input of the pose net. For all experiments, the input image is in the resolution of 416×128 if not specified.

4) *Hyperparameters and Data Augmentation*: ADAM optimizer [54] is used for training. Besides, following Bian's work [5], learning rate, α , β , and γ are set as 10^{-4} , 0.85, 0.15, and 0.1, respectively. s is set as 10. The epoch size and batch size are 2000 and 4, respectively. For each experiment, the model is trained in 500 epochs. Simultaneously, data is also augmented with random cropping, scaling, and horizontal flips. As well as color augmentations, random brightness,

TABLE I

THE DECODER ARCHITECTURE OF DEPTH NET. F0, F1, F2, F3, AND F4 ARE MULTI-SCALE FEATURES FROM THE ENCODER. $\uparrow$ IS A $2\times$ NEAREST-NEIGHBOR UPSAMPLING OF THE LAYER. GE AND CA STAND FOR GAUSSIAN ESTIMATOR AND CORRELATION-AWARE STRUCTURE, AND BOTH OF THEM ARE OPTIONAL. THE ACTIVATION FUNCTION ADOPTS RELU

Layers	Kernel size	Stride	Channel	Input
upconv1	3	2	256	F4
iconv1	3	1	256	upconv1, F3
upconv2	3	2	128	iconv1
iconv2	3	1	128	upconv2, F2
upconv3	3	2	64	iconv2
iconv3	3	1	64	upconv3, F1
upconv4	3	2	32	iconv3
iconv4	3	1	32	upconv4, F0
upconv5	3	2	16	iconv4
iconv5	3	1	64	iconv3, $\uparrow$ disp4
iconv6	3	1	32	iconv4, $\uparrow$ disp3
iconv7	3	1	16	upconv5, $\uparrow$ disp2
disp4	3	1	2	iconv2
disp3	3	1	2	iconv5
disp2	3	1	2	iconv6
disp1	3	1	2	iconv7
CA	-	-	-	disp1 or None
GE	-	-	-	CA or disp1 or None

TABLE II

THE ARCHITECTURE OF POSE NET. THE ACTIVATION FUNCTION ADOPTS RELU

Layer	Kernel size	Stride	Channel	Input
conv1	7	2	16	images
conv2	5	2	32	conv1
conv3	3	2	64	conv2
conv4	3	2	128	conv3
conv5	3	2	256	conv4
conv6	3	2	256	conv5
conv7	3	2	256	conv6
pose	1	1	14	conv7

contrast, saturation, and hue jitter with respective ranges of ± 0.2 , ± 0.2 , ± 0.2 , and ± 0.1 are adopted with a 50 percent chance. Importantly, color augmentations are merely used to images fed to the network, not to those applied to compute loss [36]. The metrics of evaluation are consistent with Zhou's work [4]. Of note is that the pre-trained weights on ImageNet are loaded to initialize the parameters of our depth net encoder for all experiments.

B. Comparisons and Results

As Table III showed, our method achieves competitive performance compared with the state-of-the-art self-supervised depth estimation method. Table IV demonstrates that our approach has state-of-the-art performance in visual odometry. Besides, we also can find that the proposed model trained on KITTI [1] can generalize to the Make3D dataset [2] unseen during training very well from Table V.

As Fig. 4 illustrated, our method also has appealing qualitative consequences on the KITTI dataset [1], especially in reflective and color-saturated regions. Fig. 5 and 6 show the qualitative results on Make3D [2] and Cityscapes [3], respectively. Of note is that the model is trained on KITTI [1], but directly applied to Make3D [2] and Cityscapes [3] without any refinements. Thus, We can learn that the proposed model

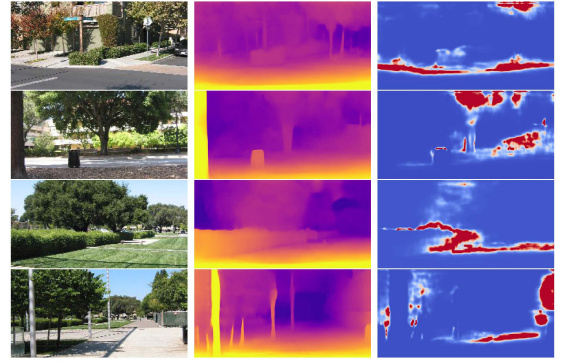


Fig. 5. Qualitative results on the Make3D dataset [2]. The first column: RGB images. The second column: predicted depth maps. The third column: predicted uncertainty maps.

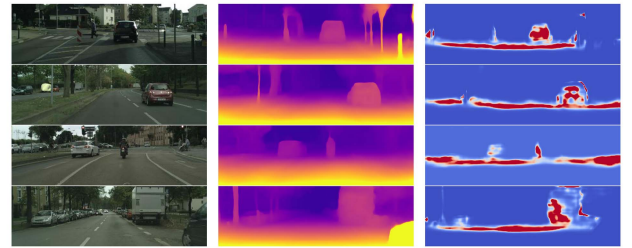


Fig. 6. Qualitative results on the Cityscapes dataset [3]. The first column: RGB images. The second column: predicted depth maps. The third column: predicted uncertainty maps.

can successfully generalize to various outdoor environments, giving promising consequences.

Notably, our model can capture pixel-wise uncertainty, helping us discover potential issues and improve performance. It can also help the decision-making system make safer decisions, thereby protecting life and property.

Besides, we can learn that high-uncertainty areas mainly exist on object edges, smooth roads, and low-texture regions of moving objects. Namely, these regions are still challenging to predict, even though some strategies have been applied. For unseen data, new appearing objects and scenes have low prediction confidence, which provides a new way of minimizing the uncertainty map to solve domain adaptation. However, some regions are high-uncertainty from our model, but it is challenging to infer the exact reasons for causing that by our simple observations. Thus, some in-depth research is needed to explore how the model reasons in the future.

C. Ablation Studies

To benefit from our method as much as possible, we carefully choose parameters for each part of our model. Also, we exhaustively do ablation studies to validate the proposed model.

1) *Correlation-Aware Structure*: We experiment with various correlation-aware units: Basic RNN, LSTM [45], and GRU [46]. As Table VI displayed, the basic RNN unit results in approximately the same consequence compared with results without correlation-aware unit, while the LSTM unit degenerates the performance significantly. In contrast, apparent performance improvements derive from the GRU unit, and

TABLE III

QUANTITATIVE RESULTS OF DEPTH ESTIMATION ON THE KITTI DATASET [1]. WE DIVIDE THE METHODS INTO TWO CATEGORIES ACCORDING TO IF TRUE DEPTHS SUPERVISE THEM. IN EACH CATEGORY, THE BEST PERFORMANCES ARE MARKED **BOLD**, AND THE SECOND-BEST PERFORMANCES ARE MARKED UNDERLINE IN THE SELF-SUPERVISED BLOCK. (B) STANDS FOR BINOCULAR/STEREO INPUT PAIRS, AS (J) DENOTES JOINT LEARNING OF MULTIPLE TASKS. GT REPRESENTS GROUND TRUTH. NOTABLY, ALL SELF-SUPERVISED METHODS ARE TRAINED IN RESOLUTION 416×128 FOR A FAIR COMPARISON. † REPRESENTS THAT WE IMPLEMENT OUR METHOD UPON MONODEPTH2 [36], ADDING RESNET-18 [47] ENCODER, AUTO-MASKING AND MINIMUM REPROJECTION LOSS [36] FOR POSE AND DEPTH ESTIMATION, RESPECTIVELY

Methods	GT?	Errors ↓				Errors ↑		
		AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
Eigen et al., coarse [14]	✓	0.214	1.605	6.563	0.292	0.673	0.884	0.957
Eigen et al., fine [14]	✓	0.203	1.548	6.307	0.282	0.702	0.890	0.958
Liu et al. [48]	✓	0.202	1.614	6.523	0.275	0.678	0.895	0.965
Kuznetsov et al. (B) [41]	✓	0.113	0.741	4.621	0.189	0.862	0.960	0.986
DORN [29]	✓	0.072	0.307	2.727	0.120	0.932	0.984	0.994
SfMLearner [4]		0.208	1.768	6.856	0.283	0.678	0.885	0.957
Yang et al. (J) [49]		0.182	1.481	6.501	0.267	0.725	0.906	0.963
GeoNet-VGG (J) [33]		0.164	1.303	6.090	0.247	0.765	0.919	0.968
GeoNet-Resnet (J) [33]		0.155	1.296	5.857	0.233	0.793	0.931	0.973
DDVO [50]		0.151	1.257	5.583	0.228	0.810	0.936	0.974
SC-SfMLearner [5]		0.149	1.137	5.771	0.230	0.799	0.932	0.973
Struct2depth [37]		0.141	1.026	5.291	0.215	0.816	0.945	0.979
Monodepth2 [36]		0.128	1.087	5.171	<u>0.204</u>	0.855	<u>0.953</u>	0.978
Our		0.144	<u>0.966</u>	<u>5.078</u>	0.208	0.815	0.945	0.981
Our†		<u>0.130</u>	0.957	4.907	0.203	<u>0.851</u>	0.954	<u>0.980</u>

TABLE IV

THE ABSOLUTE TRAJECTORY ERROR (ATE) ON THE KITTI ODOMETRY DATASET [1] FOLLOWING ZHAN'S SPLIT [34]. THE BEST PERFORMANCES ARE MARKED **BOLD**

Methods	Seq. 09	Seq. 10
Mean Odometry	0.032 ± 0.026	0.028 ± 0.023
ORB-SLAM (short)	0.064 ± 0.141	0.064 ± 0.130
ORB-SLAM (full)	0.014 ± 0.008	0.012 ± 0.011
SfMLearner [4]	0.021 ± 0.017	0.020 ± 0.015
Vid2Depth [51]	0.013 ± 0.010	0.012 ± 0.011
Monodepth2 [36]	0.017 ± 0.008	0.015 ± 0.010
GeoNet [33]	0.012 ± 0.007	0.012 ± 0.009
Struct2depth [37]	0.011 ± 0.006	0.011 ± 0.010
Our (416×128)	0.011 ± 0.006	0.009 ± 0.007
Our (832×256)	0.009 ± 0.007	0.009 ± 0.007

TABLE V

QUANTITATIVE RESULTS OF DEPTH ESTIMATION ON THE MAKE3D DATASET [2]. THE BEST PERFORMANCES ARE MARKED **BOLD**

Methods	GT?	Errors ↓			
		AbsRel	SqRel	RMS	RMSlog
Karsch et al. [52]	✓	0.428	5.079	8.389	0.149
Liu et al. [53]	✓	0.475	6.562	10.05	0.165
Laina et al. [15]	✓	0.204	1.840	5.683	0.084
SfMLearner [4]		0.383	5.321	10.470	0.478
DDVO [50]		0.387	4.720	8.090	0.204
Monodepth [38]		0.544	10.94	11.760	0.193
Monodepth2 [36]		0.322	3.589	7.417	0.163
Our		0.301	3.143	6.972	0.351

thus GRU unit is chosen by us. Furthermore, we explore the influence caused by the number of GRU layer. We can see that there is no performance improvement with increasing the number of GRU layers from Table VII. Inversely, the performance is seemingly going backward along with the increasing

number of GRU gradually. We guess that is because multi GRU is difficult to converge, or the constraint of multi GRU is too strong to predict a reasonable depth. Comprehensively, we determine our correlation-aware structure as a single GRU layer.

2) *CLAEH*: We experiment with different configurations to discover the best parameters. Table VIII shows that strategy 2 performs a little bit better than strategy 1, but whichever can significantly improve performance. Table IX indicates that there is no performance improvement by increasing the threshold. We speculate that a large threshold may dramatically change the color and amplify the noise, thus inhibiting performance progress. Note that the default grid size of CLAEH is adopted for all experiments. As Fig. 3 illustrated, the CLAEH method can significantly improve the gradient locality issue, particularly in smooth roads and shadows. Comprehensively, we determine our data preprocessing as the second way with threshold 5.

3) *Solidity-Aware Mask*: Our solidity-aware mask is to pick the low-uncertainty predictions out and overlook the high-uncertainty predictions, thereby mitigating the tight issues' influence. Here, we assume that potential problematic regions cause high uncertainties. To benefit from it as much as possible, we experiment with different kernel sizes, as Table X showed. It turns out that small kernel size performs better than large ones. Thus, we determine the receptive field of our solidity-aware mask as 3×3 .

4) *Fusion Scaling*: As Table XI showed, the frequently-used Median method [4] prefers to result in better AbsRel, while the Mean method prefers to obtain better RMS. Our fusion strategy can tradeoff between Mean and Median, even having better performances on other indexes.

5) *Overall*: In this part, we validate each part of our method by adopting the most suitable configurations from previous

TABLE VI

ABLATION STUDIES ON DIFFERENT CORRELATION-AWARE UNITS. - DENOTES THE BASELINE EXPERIMENT WITHOUT CORRELATION-AWARE UNITS. THE BEST PERFORMANCES ARE MARKED **BOLD**

Units	Errors ↓				Errors ↑		
	AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
-	0.164	1.170	5.550	0.236	0.774	0.925	0.972
RNN	0.165	1.146	5.753	0.239	0.772	0.923	0.972
LSTM	0.184	1.445	6.719	0.274	0.723	0.894	0.958
GRU	0.154	1.100	5.536	0.228	0.787	0.932	0.975

TABLE VII

ABLATION STUDIES ON DIFFERENT NUMBER OF GRU LAYER. - DENOTES THE BASELINE EXPERIMENT WITHOUT CORRELATION-AWARE UNITS. THE BEST PERFORMANCES ARE MARKED **BOLD**

Layer number(s)	Errors ↓				Errors ↑		
	AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
-	0.164	1.170	5.550	0.236	0.774	0.925	0.972
1	0.154	1.100	5.536	0.228	0.787	0.932	0.975
2	0.163	1.103	5.592	0.236	0.777	0.926	0.972
3	0.168	1.260	5.776	0.242	0.771	0.922	0.969

TABLE VIII

THE ABLATION STUDIES ON DIFFERENT DATA PREPROCESSING STRATEGIES. - DENOTES THE BASELINE EXPERIMENT WITHOUT DATA PREPROCESSING. THE BEST PERFORMANCES ARE MARKED **BOLD**

Strategies	Errors ↓				Errors ↑		
	AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
-	0.154	1.087	5.443	0.226	0.791	0.932	0.975
1	0.150	1.042	5.313	0.219	0.797	0.938	0.978
2	0.151	1.049	5.246	0.217	0.803	0.939	0.978

TABLE IX

THE ABLATION STUDIES ON DIFFERENT THRESHOLDS OF STRATEGY TWO. - DENOTES THE BASELINE EXPERIMENT WITHOUT DATA PREPROCESSING. THE BEST PERFORMANCES ARE MARKED **BOLD**

Thresholds	Errors ↓				Errors ↑		
	AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
-	0.154	1.087	5.443	0.226	0.791	0.932	0.975
5	0.151	1.049	5.246	0.217	0.803	0.939	0.978
15	0.151	1.023	5.323	0.219	0.798	0.937	0.978
25	0.153	1.048	5.386	0.222	0.793	0.934	0.976

TABLE X

THE ABLATION STUDIES ON VARIOUS KERNEL SIZES. - DENOTES THE BASELINE EXPERIMENT WITHOUT SOLIDITY-AWARE MASK. - DENOTES THE BASELINE EXPERIMENT WITHOUT SOLIDITY-AWARE MASK. THE BEST PERFORMANCES ARE MARKED **BOLD**

Kernel size	Errors ↓				Errors ↑		
	AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
-	0.151	1.049	5.246	0.217	0.803	0.939	0.978
3×3	0.142	0.985	5.218	0.212	0.814	0.942	0.980
7×7	0.148	1.011	5.301	0.219	0.802	0.939	0.978
11×11	0.151	1.077	5.219	0.217	0.804	0.939	0.978

TABLE XI

THE ABLATION STUDIES ON DIFFERENT SCALING STRATEGIES. THE BEST PERFORMANCES ARE MARKED **BOLD**

Strategies	Errors ↓				Errors ↑		
	AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
Median	0.142	0.985	5.218	0.212	0.814	0.942	0.980
Mean	0.153	0.990	5.005	0.211	0.803	0.944	0.981
Fusion	0.144	0.966	5.078	0.208	0.815	0.945	0.981

ablation studies. As Table XII showed, each part of our proposed method can obtain performance improvement, especially for our correlation-aware structure, Gaussian estimator, CLAHE strategy, and solidity-aware mask. Note that we write the combination of photometric loss and SSIM loss as the baseline for simplicity.

Additionally, we also apply our method to higher resolution images (832 × 256), and the quantitative results are shown in Table XIII. Note that the methods exhibiting in Table XIII are trained in the same resolution for a fair comparison. It turns out that our model can achieve better results with a higher resolution.

TABLE XII

ABLATION STUDIES ON PROPOSING METHODS. B, S, CA, G, C, M, AND F DENOTE BASELINE, EDGE-AWARE SMOOTHNESS, CORRELATION-AWARE STRUCTURE, GAUSSIAN ESTIMATOR, CLAHE STRATEGY, SOLIDITY-AWARE MASK, AND FUSION SCALING METHOD, RESPECTIVELY. THE BEST PERFORMANCES ARE MARKED **BOLD**

Methods	Errors ↓				Errors ↑		
	AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
B	0.165	1.315	5.986	0.241	0.775	0.922	0.970
B+S	0.164	1.170	5.550	0.236	0.774	0.925	0.972
B+S+CA	0.154	1.100	5.536	0.228	0.787	0.931	0.975
B+S+CA+G	0.154	1.087	5.443	0.226	0.791	0.932	0.975
B+S+CA+G+C	0.151	1.049	5.246	0.217	0.803	0.939	0.978
B+S+CA+G+C+M	0.142	0.985	5.218	0.212	0.814	0.942	0.980
B+S+CA+G+C+M+F	0.144	0.966	5.078	0.208	0.815	0.945	0.981

TABLE XIII

QUANTITATIVE RESULTS OF OUR METHOD IN HIGHER RESOLUTION IMAGES (832×256). THE BEST PERFORMANCES ARE MARKED **BOLD**. (J) DENOTES JOINT LEARNING OF MULTIPLE TASKS

Methods	Errors ↓				Errors ↑		
	AbsRel	SqRel	RMS	RMSlog	< 1.25	< 1.25 ²	< 1.25 ³
EPC++ [55]	0.141	1.029	5.350	0.216	0.816	0.941	0.976
CC (J) [32]	0.140	1.070	5.326	0.217	0.826	0.941	0.975
SC-SIMLearner [5]	0.137	1.089	5.439	0.217	0.830	0.942	0.975
Our	0.136	0.895	4.834	0.199	0.832	0.950	0.982

TABLE XIV

MODEL SIZE AND INFERENCE TIME

items	Depth net		Pose net	
	416×128	832×256	416×126	832×256
Inference time	19.193 ms	33.259 ms	3.825 ms	3.862 ms
Parameter numbers	57.6 M		6.37 M	

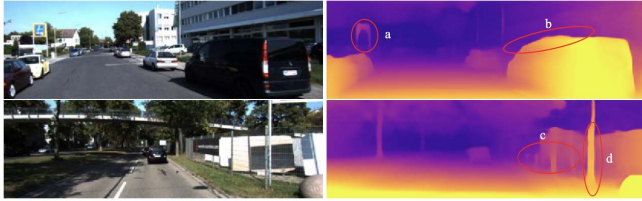


Fig. 7. Failure cases of our model. The first column: RGB images. The second column: predicted depth maps.

6) *Model Size and Inference Time*: Table XIV shows the model size and inference time of our model in different resolutions. Of note is that our model runs on a single 1080 GPU card, and the inference time is averaged over 100 iterations. Besides, it takes about two days to train our model with input images in resolution of 416×128 , while three days for input images in resolution of 832×256 . Besides, we can find that it takes 19.193 ms and 33.259 ms for a single image's depth estimation in resolutions of 416×128 and 832×256 , respectively, meeting real-time perfectly. The pose estimation speed is over 250 sequences per second.

VI. CONCLUSION

This paper proposes an innovative method to model the underlying relations between depths and predict uncertainty, benefiting performance improvement. The CLAHE and solidity-aware mask strategies are also proposed to mitigate

the high-uncertainty regions' influence, further improving the performance. The experiments indicate that our model achieves competitive performance compared with the state-of-the-art methods. Besides, our model has strong generalization capability and can generalize to various outdoor environments, even low-quality real-world data.

VII. LIMITATIONS & FUTURE WORK

As shown in Figure 7, the current model has several limitations as follows.

- Our model does not have any shape and smoothness priors, resulting in some unexpected failure cases when predicting the regular objects, such as the brand board (circle a) and car's edges (circle b). It is unsurprising since we do not require the network to learn this advanced semantics knowledge. However, necessary priors can significantly benefit performance improvement.
- Our model is challenging to capture very fine things (circle c, iron gauze) and distinguish too close objects (circle d, two steel poles). They also require that the network have an in-depth semantic understanding of the scene.

In the future, we will work on these issues to further improve the performance.

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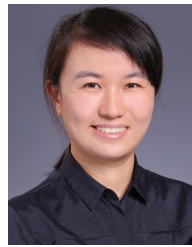
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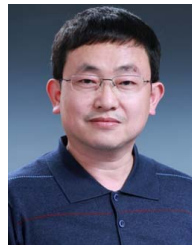
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